

Axioms of Algebra

- **Closure Law:**

- For Addition: $a + b \in \mathbb{R}$ for all $a, b \in \mathbb{R}$.
- For Multiplication: $a \cdot b \in \mathbb{R}$ for all $a, b \in \mathbb{R}$.

- **Commutative Law:**

- For Addition: $a + b = b + a$.
- For Multiplication: $a \cdot b = b \cdot a$.

- **Associative Law:**

- For Addition: $(a + b) + c = a + (b + c)$.
- For Multiplication: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

- **Distributive Law:**

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

- **Identity Law:**

- For Addition: $a + 0 = a$.
- For Multiplication: $a \cdot 1 = a$.

- **Inverse Law:**

- For Addition: $a + (-a) = 0$.
- For Multiplication: $a \cdot \frac{1}{a} = 1$ (if $a \neq 0$).